

Solutions to Exercise Sheet 2

Exercise 1: Non-additivity of QPC Resistances in Series

1. Landauer-Büttiker formula for multi-terminal setup:

The Landauer-Büttiker formalism for multi-terminal conductors reads:

$$\left(\frac{h}{2e^2}\right) I_i = (N_i - R_{ii})V_i - \sum_{j \neq i} T_{ij} V_j$$

with N_i the number of modes at terminal i , R_{ii} the reflection probability at contact i , and T_{ij} the transmission probability from j to i .

Current conservation requires:

$$N_i - R_{ii} = \sum_{j \neq i} T_{ji}$$

To obtain the two-terminal conductance, set current to flow from source (V_1) to drain ($V_3 = 0$), while V_2, V_4 are internal leads with no net current ($I_2 = I_4 = 0$). The system of equations can then be solved for $I_1 = -I_3$ in terms of V_1 , yielding the total conductance G .

2. Large separation of QPCs: Ohmic addition in zero magnetic field ($T_F \approx 0$, $T_L = T_R$):

With large separation, direct transmission between 1 and 3 (and 2 and 4) is negligible: $T_{13} = T_{31} = T_F \approx 0$, and time-reversal symmetry gives $T_L = T_R$.

For each QPC, the conductance is quantized: $G_{\text{QPC}} = N(2e^2/h)$. In series, classical Ohmic addition yields

$$R_{\text{tot}} = R_1 + R_2 \implies G_{\text{tot}}^{-1} = G_1^{-1} + G_2^{-1}$$

For two identical QPCs,

$$G = \frac{G_{\text{QPC}}}{2} = \frac{N(2e^2/h)}{2}$$

This matches the expected result for incoherent transport with large separation in zero field.

3. Short separation, zero magnetic field:

If the QPCs are close, electrons cannot scatter between them, and the system acts as one single QPC. The only limiting conductance is set by the QPC with the smallest number of modes.

Thus,

$$G = N(2e^2/h)$$

where $N = \min(N_1, N_2)$, i.e., the QPC with fewer open channels determines the conductance. There is no Ohmic addition: the two QPCs behave as one.

4. Strong magnetic field ($T_F, T_L \ll T_R$):

For a large magnetic field, cyclotron orbits dominate, and direct transmission (T_F) and left transmission (T_L) vanish: $T_F, T_L \approx 0$, $R_{RR} \approx R_{LL} \approx 0$. Solving the Landauer-Büttiker equations in this regime, one finds:

$$G = \frac{2e^2}{h} \left(\frac{2}{N_{\text{QPC}}} - \frac{1}{N_{2\text{DEG}}} \right)^{-1}$$

where N_{QPC} is the number of modes in the QPCs and $N_{2\text{DEG}}$ is the number of edge channels in the 2DEG region between the QPCs (i.e., the Landau level degeneracy).

5. Origin of the 'camel back' in $G(B)$:

In strong B , the 2DEG forms Landau levels, and the number of edge channels in the 2DEG is

$$N_{2\text{DEG}} = \frac{1}{2} k_F l_c$$

where $l_c = \hbar k_F / (eB)$ is the cyclotron radius.

As B increases, l_c decreases, and $N_{2\text{DEG}}$ decreases (number of occupied Landau levels drops). According to the formula above, G increases as $N_{2\text{DEG}}$ decreases. However, once the cyclotron diameter ($2l_c$) becomes smaller than the QPC width ($2l_c < W_{\text{QPC}}$), the QPC itself supports fewer edge states (Landau levels quantized in the QPC). Now N_{QPC} starts to decrease as well, reducing G .

Thus, G first increases (due to reduced $N_{2\text{DEG}}$), then decreases (as N_{QPC} becomes a bottleneck). The 'camel back' peak occurs when $2l_c \sim W_{\text{QPC}}$: at this point, both $N_{2\text{DEG}}$ and N_{QPC} compete, causing a maximum in G .

Exercise 2: Interferences in a Ring

1. Estimate the period ΔB of Aharonov-Bohm oscillations and deduce the ring diameter:

The flux quantum is $\Phi_0 = h/e$. The period in magnetic field ΔB satisfies

$$\Delta B \cdot A = \Phi_0 \implies \Delta B = \frac{h}{eA}$$

where A is the area enclosed by the ring.

Suppose (from Fig. 2) the observed AB period is, e.g., $\Delta B \approx 15$ mT (read the value from the figure, or as specified in the sheet).

Then,

$$A = \frac{h}{e\Delta B} = \frac{6.626 \times 10^{-34} \text{ Js}}{1.6 \times 10^{-19} \text{ C} \cdot 15 \times 10^{-3} \text{ T}} \approx 2.76 \times 10^{-12} \text{ m}^2$$

Assuming a circular ring, $A = \pi r^2$, so $r = \sqrt{A/\pi} \approx 29.7 \text{ } \mu\text{m}$, diameter $d = 2r \approx 59.4 \text{ } \mu\text{m}$.

2. Thick ring:

If the ring is thick, with inner diameter $d_{in} = 1 \text{ } \mu\text{m}$ and outer $d_{out} = 2 \text{ } \mu\text{m}$, the width is comparable to the mean free path and the electron wavelength.

The Aharonov-Bohm effect relies on well-defined single-valued phase accumulated by electrons traversing the ring. In a thick ring, the possible electron paths enclose a range of areas, leading to a range of phase shifts. The interference is washed out due to averaging over many different paths, and observable AB oscillations disappear.

This is analogous to the loss of interference in optics when the slit width exceeds the coherence length: it is similar to the 'white light fringe effect'.

3. A-B oscillations in a (Bi,Sb)Te nanowire:

In the experiment of Rößler et al., oscillations in magnetoresistance as a function of B are observed.

From the period ΔB of the oscillations (read off from Fig. 3, or as reported, e.g. $\Delta B \approx 0.19$ T), we infer the enclosed area

$$A = \frac{h}{e\Delta B}$$

Suppose $\Delta B = 0.19$ T,

$$A = \frac{6.626 \times 10^{-34} \text{ Js}}{1.6 \times 10^{-19} \text{ C} \cdot 0.19 \text{ T}} \approx 2.18 \times 10^{-14} \text{ m}^2$$

The magnetic field is applied parallel to the axis of the nanowire (axial field), so A is the projected cross-sectional area, $A = w \times t$, where w is the width and t the thickness of the nanowire. With $w = 150 \text{ nm}$, $t = 30 \text{ nm}$:

$$A_{\text{geom}} = (150 \times 10^{-9})(30 \times 10^{-9}) = 4.5 \times 10^{-15} \text{ m}^2$$

which is smaller than the value extracted from AB period.

The difference can be explained: the experimentally extracted area may be larger because (i) the effective electron path encloses a larger area (e.g. due to penetration of wavefunction into the oxide or wider conduction channel), (ii) flux focusing due to finite penetration depth or (iii) surface states extending beyond the geometric boundaries.